

PocketSmart AI: Your Smart Budget & Recommendation Assistant

Project Description

PocketSmart AI uses Generative AI to turn a budget into a personalised, shop-able plan for **home interiors, parties and jewelry**. A FastAPI backend sends validated user input (text, numbers and an optional outfit image) to Google Gemini, then re-checks the answer so the plan never exceeds the budget, and attaches search links for Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO and other platforms. Users register, sign in, and revisit saved plans from their history.

Scenarios

Scenario 1: Home Interior Planning. The user opens the Home Budget Planner, enters a budget, quantities of lights, ceiling fans, furniture and dining tables, and selects rooms. Gemini recommends cost-effective options for each category across IKEA, Amazon, Flipkart, Myntra and Ajio, balanced for function, style and price.

Scenario 2: AI-Based Party Budget Planning. In the Party Planner the user enters budget, guest count, party type and venue. The budget is allocated across venue, catering, decoration and entertainment, with links to Swiggy, Zomato, BookMyShow, OYO and more, plus a contingency amount.

Scenario 3: Jewelry Recommendations. The Jewelry Planner takes a budget, occasion and style preferences, and optionally an outfit photo. Gemini analyses colours and formality and suggests matching pieces from Amazon, Flipkart, BlueStone, Tanishq, CaratLane and others.

Technical Architecture

Three tiers: a responsive HTML/CSS/JS front end (Jinja2 templates), a FastAPI back end (routes, JWT sessions, validation, budget engine) and external services (Gemini API, shop search URLs) with SQLite for storage. See 3. *Project Design Phase / Solution Architecture* for the diagram.

Pre-requisites

- 1. Python 3.10+ - <https://www.python.org>
- 2. FastAPI - <https://fastapi.tiangolo.com>
- 3. Google AI Studio account and Gemini API key - <https://aistudio.google.com/apikey>
- 4. Git and a GitHub account
- 5. Basic HTML, CSS and JavaScript

Project Workflow

Milestone	Activities	Where in the code
1. Gemini AI Initialization	1.1 Create Google AI Studio account; 1.2 generate and store the API key in <code>.env</code> ; 1.3 validate text and image connectivity	<code>.env.example</code> , <code>config.py</code> , <code>check_gemini.py</code>

Milestone	Activities	Where in the code
2. Core Functionalities Development	2.1 FastAPI initialization and imports; 2.2 planner endpoints with Gemini prompts; 2.3 login/register routing; 2.4 modular structure with /token, /session-info, /session-data	main.py, gemini_utils.py, auth.py, models.py
3. Backend - FastAPI Integration	3.1 /home-budget, /party-budget, /jewelry-budget (aliases /generate-*) and auth routes; 3.2 /recommendation-details/{id}; 3.3 CORS, /history; 3.4 startup task and __main__	main.py
4. UI Development	4.1 landing page, forms, card-style results; dynamic rendering with JavaScript	templates/, static/
5. Testing & Optimization	5.1 real-world inputs across the planners; 5.2 response quality and budget adherence; 5.3 prompt tuning and validation; 5.4 fallback recommendations	tests/, gemini_utils.py

Milestone 1: Gemini AI Initialization

1. Sign in at <https://aistudio.google.com> and choose **Get API key** then **Create API key**.
2. Copy the key into .env as GOOGLE_API_KEY (never commit .env).
3. Run `python check_gemini.py`. It sends a text prompt and a text + image prompt, tries the configured model and then fallback models, and prints which model to use.

Note: the original brief refers to "Gemini 1.5 Flash Pro". That model name does not exist and the 1.5 family is retired, so the model is configurable (GEMINI_MODEL).

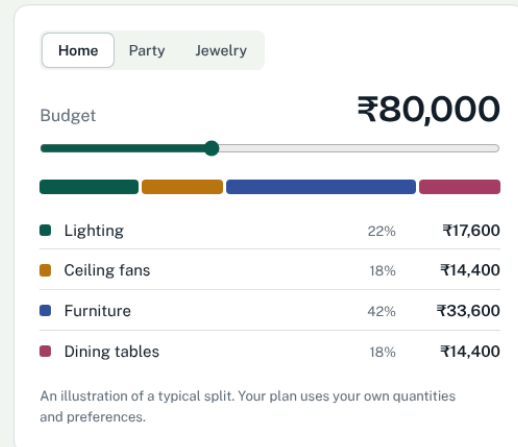
Milestone 2 and 3: Backend

- `config.py` loads settings; `models.py` validates inputs; `database.py` stores users and plans; `auth.py` handles bcrypt, JWT and sessions.
- `gemini_utils.py` builds one prompt per planner, calls Gemini (JSON mode), sanitises the reply, recomputes totals, trims over-budget items, builds shopping links and falls back to default plans if needed.
- `main.py` exposes the routes: /register, /login, /token, /logout, /session-info, /session-data, /home-planner, /party-planner, /jewelry-planner, /home-budget, /party-budget, /jewelry-budget, /recommendation-history, /recommendation-details/{id}, /history, /dashboard, /health. Interactive API docs are at /docs.

Milestone 4: UI Development

Know where every rupee goes before you spend it.

Enter a budget for your home, a party or a piece of jewelry. PocketSmart splits it across what you need, checks the total, and links each item to the stores that sell it.

[Create a free account](#)[Sign in](#)

Three planners, one account

Each planner asks only for what it needs.

Home interior

Lights, ceiling fans, furniture and dining tables for the rooms you choose, with options from IKEA, Amazon, Flipkart, Myntra and Ajo.

Budget, quantities, rooms

Party

Venue, catering, decoration and entertainment for your guest count, with a contingency buffer. Links to Swiggy, Zomato, BookMyShow, OYO and more.

Budget, guests, event, venue

Jewelry

Pieces matched to the occasion and, if you add a photo, to the colours and style of your outfit. Links to BlueStone, Tanishq, CaratLane, Amazon and more.

Budget, occasion, optional photo

How it works

1

Enter your budget

Choose a planner and fill in a short form. Quick amounts save typing.

2

Get a checked plan

Gemini proposes the items. The server rechecks every price and removes anything that would take you over budget.

3

Shop and come back

Open the store links for each item. Every plan is saved in your history.

The AI suggests. The server verifies.

Totals you can trust

Planned spend and remaining budget are calculated by the server, never taken from the AI's own sums.

Always an answer

If the AI service is unavailable, you get a clearly labelled standard estimate instead of an error.

Your plans stay yours

Passwords are hashed, sessions expire after 30 minutes, and only you can open your saved plans.

Create your account

Save your plans and come back to them any time.

Username

3 to 30 characters: letters, numbers, dot, dash or underscore.

Email

Password

[Show](#)

Use at least 8 characters.

Confirm password

Create account

Already have an account? [Sign in](#)

Sign in

Welcome back. Pick up where you left off.

Account created. Sign in to start planning.

Username

Password

Show

Sign in

New to PocketSmart? [Create an account](#)

PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.

Hello, demo_091d44

What would you like to plan today?

Start a plan

Home interior

Lights, fans, furniture and tables for your rooms.

Open

Party

Venue, catering, decoration and entertainment.

Open

Jewelry

Pieces for the occasion, matched to your outfit.

Open

Recent plans

● Jewelry

3 Oct 2026, 4:48 pm

₹20,000

Standard estimate

● Party

3 Oct 2026, 4:48 pm – Catering, Decoration, Entertainment, Contingency, Venue

₹40,000

Standard estimate

● Home interior

3 Oct 2026, 4:48 pm – Lighting, Ceiling fans, Furniture, Dining tables, Living room, Kitchen

₹80,000

Standard estimate

[See all plans](#)

PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.

Milestone 5: Testing & Optimization

Home Interior Planner and recommendations

Tell us your budget and what the home needs. We split the money across lighting, fans, furniture and tables.

Home interior plan			Print plan
Saved to your history just now.			
The AI service was unavailable, so this is a standard estimate based on typical prices. Try again later for a tailored plan.			
Budget	Planned	Left over	
₹80,000	₹42,950	₹37,050	
Lighting		3%	₹2,250
Ceiling fans		10%	₹7,600
Furniture		6%	₹4,800
Dining tables		16%	₹12,500
Living room		18%	₹14,000
Kitchen		2%	₹1,800
Unspent		46%	₹37,050
Lighting			₹2,250 of ₹2,250
LED batten / panel light			₹2,250
Energy-efficient 20W LED light, cool or warm white.			5 × ₹450
Search on	Amazon	Flipkart	IKEA Myntra Ajo
Ceiling fans			₹7,600 of ₹7,600
BLDC ceiling fan 1200mm			₹7,600
Energy-saving BLDC ceiling fan with remote.			2 × ₹3,800
Search on	Amazon	Flipkart	IKEA Myntra Ajo
Furniture			₹4,800 of ₹4,800
Engineered-wood chair			₹4,800
Sturdy, easy-to-assemble chair for living areas.			2 × ₹2,400
Search on	Amazon	Flipkart	IKEA Myntra Ajo
Dining tables			₹12,500 of ₹12,500
4-seater dining table set			₹12,500
Compact 4-seater dining table with chairs.			
Search on	Amazon	Flipkart	IKEA Myntra Ajo
Living room			₹14,000 of ₹14,000
3-seater fabric sofa			₹14,000
Comfortable sofa for the living room.			
Search on	Amazon	Flipkart	IKEA Myntra Ajo
Kitchen			₹1,800 of ₹1,800
Modular kitchen storage rack			₹1,800
Stainless-steel rack for utensils and spices.			
Search on	Amazon	Flipkart	IKEA Myntra Ajo

Subscription

Partnership

Enterprise

Team

Agency

Other

Tips

Offline mode: these are typical-price estimates, not live AI picks.

Compare prices across platforms and watch for festive-season sales.

PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.

Party Planner and recommendations

Party planner

Give us the budget, guest count and event. We divide it between venue, catering, decoration and entertainment, and keep a small buffer.

Budget and guests

Total budget

₹ 40,000

₹20,000

₹50,000

₹1,00,000

Number of guests

30

Event

Party type

Birthday

Venue

Banquet hall

What to plan for

Catering

Decoration

Entertainment

Preferences

Anything else we should know

Vegetarian menu, a jungle theme, music for all ages

Build my plan

Party plan

Print plan

Saved to your history just now.

The AI service was unavailable, so this is a standard estimate based on typical prices. Try again later for a tailored plan.

Budget

₹40,000

Planned

₹37,997

Left over

₹2,003

Catering

36% ₹14,490

Decoration

14% ₹5,796

Entertainment

14% ₹5,474

Contingency

6% ₹2,576

Venue

24% ₹9,661

Unspent

5% ₹2,003

Catering

₹14,490 of ₹14,490

Birthday - Catering per guest

₹14,490

Veg meal / snacks per guest via Swiggy or Zomato party packs.

30 × ₹483

Search on

Swiggy

Zomato

Decoration

₹5,796 of ₹5,796

Birthday - Decoration kit

₹5,796

Balloons, banners, lights and table decor for the theme.

Search on

Amazon

Flipkart

Meesho

Myntra

Entertainment

₹5,474 of ₹5,474

Birthday - Entertainment and games

₹5,474

Music playlist, speaker rental and group games.

Search on

BookMyShow

Amazon

Flipkart

Contingency

₹2,576 of ₹2,576

Birthday - Contingency buffer

₹2,576

Reserve for last-minute needs.

Search on

Amazon

Flipkart

Google

Venue

₹9,661 of ₹9,661

Birthday - Venue hire

₹9,661

Banquet hall or terrace space for your guest count.

Search on

Google

Booking

MakeMyTrip

OYO

NoBroker

Venue ideas

Birthday party hall

₹9,661

Banquet hall. up to 30 guests

Search on

Google

Booking

MakeMyTrip

OYO

NoBroker

Tips

- Offline mode: allocation follows a standard party-budget split.
- Confirm vendor quotes before booking.

PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.

Jewelry Planner (with outfit image) and recommendations

PocketSmart

Dashboard

Home

Party

Jewelry

History

Log out

Jewelry planner

Pick an occasion and a budget. Add a photo of your outfit and the AI matches jewelry to its colours and style.

Budget

Total budget

₹ 20,000

₹10,000 ₹25,000 ₹50,000

Occasion and style

Occasion

Wedding

Style preferences

Gold, traditional

Outfit photo (optional)

_outfit.png (2 KB). Choose a different photo to replace it.

Remove photo

Build my plan

Jewelry plan

Print plan

Saved to your history just now.

The AI service was unavailable, so this is a standard estimate based on typical prices. Try again later for a tailored plan.

Budget

Planned

Left over

₹20,000

₹19,000

₹1,000

Recommended pieces

Necklace

₹9,000

A traditional gold-plated necklace suited to a wedding. traditional gold-plated

Search on Amazon Flipkart BlueStone Tanishq CaratLane Melorra Meesho

Earrings

₹6,000

A traditional gold-plated earrings suited to a wedding. traditional gold-plated

Search on Amazon Flipkart BlueStone Tanishq CaratLane Melorra Meesho

Bracelet

₹4,000

A traditional gold-plated bracelet suited to a wedding. traditional gold-plated

Search on Amazon Flipkart BlueStone Tanishq CaratLane Melorra Meesho

Tips

Offline mode: outfit photo analysis needs the Gemini service.

Match metal tone (gold/silver) across all pieces.

PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.

Input validation

Total budget

₹ 5,000

₹20,000₹50,000₹1,00,000

Number of guests

0

Enter a guest count from 1 to 5,000.

Event

Party type

Venue

Birthday▼Home▼

What to plan for

CateringDecorationEntertainment

Preferences

Anything else we should know

Vegetarian menu, a jungle theme, music for all ages

Build my plan

Your plan appears here

Set your budget and guest count, then choose Build my plan.

• Every total is checked on the server, so the plan never goes over your budget.

• Each item links to search results on the right stores.

• Plans are saved to your history.

PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.

Recommendation history and details

Your plans

Every plan you build is saved here. Open one to see the full breakdown and store links.

- All
- Home
- Party
- Jewelry

Jewelry 3 Oct 2026, 4:48 pm	₹20,000 Standard estimate
Party 3 Oct 2026, 4:48 pm – Catering, Decoration, Entertainment, Contingency, Venue	₹40,000 Standard estimate
Home interior 3 Oct 2026, 4:48 pm – Lighting, Ceiling fans, Furniture, Dining tables, Living room, Kitchen	₹80,000 Standard estimate

PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.

Your plans

Every plan you build is saved here. Open one to see the full breakdown and store links.

- All
- Home
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Home interior 3 Oct 2026, 4:48 pm – Lighting, Ceiling fans, Furniture, Dining tables, Living room, Kitchen	₹80,000 Standard estimate

PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.

Jewelry plan

Saved 3 Oct 2026, 4:48 pm

The AI service was unavailable, so this is a standard estimate based on typical prices. Try again later for a tailored plan.

Budget	Planned	Left over
₹20,000	₹19,000	₹1,000

Recommended pieces

Necklace A traditional gold-plated necklace suited to a wedding. traditional gold-plated Search on Amazon Flipkart BlueStone Tanishq CaratLane Melorra Meesho	₹9,000
Earrings A traditional gold-plated earrings suited to a wedding. traditional gold-plated Search on Amazon Flipkart BlueStone Tanishq CaratLane Melorra Meesho	₹6,000
Bracelet A traditional gold-plated bracelet suited to a wedding. traditional gold-plated Search on Amazon Flipkart BlueStone Tanishq CaratLane Melorra Meesho	₹4,000

Tips

- Offline mode: outfit photo analysis needs the Gemini service.
- Match metal tone (gold/silver) across all pieces.

These screenshots were captured by the automated browser test `tools/capture_screenshots.py` running **in offline-fallback mode** (no API key), which is why results are labelled "Offline estimate". With a Gemini key the label reads "Gemini AI" and the items come from the model.

Conclusion

PocketSmart AI combines FastAPI with Google Gemini to give budget-aware recommendations for home interiors, parties and jewelry. The server verifies every plan against the budget, falls back gracefully if the AI is unavailable, protects user data with hashed passwords and signed sessions, and links each item to the platforms where it can be bought.